

BUFFALO JOHNSON COUNTY AP, WY, USA

WMO: 726654

Lat: 44.381N

Lon: 106.721W

Elev: 1514

StdP: 84.41

Time Zone: -7.00 (NAM)

Period: 99-19

WBAN: 94054

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-21.2	-18.0	-25.3	0.5	-19.9	-22.4	0.6	-15.8	16.2	-4.2	14.6	-4.3	3.7	350	0.641

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.2	33.8	15.7	32.2	15.5	30.2	15.2	18.0	28.0	17.2	27.6	16.5	27.1	4.6	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.7	12.6	21.6	13.6	11.7	21.0	12.6	10.9	20.4	56.8	27.8	54.0	27.3	51.5	26.9	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.3	12.2	10.8	-25.4	36.9	3.0	1.6	-27.6	38.1	-29.3	39.0	-31.0	39.9	-33.2	41.0		
(5)			-25.7	19.9	3.0	2.1	-27.9	21.4	-29.6	22.7	-31.2	23.9	-33.4	25.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.2	-2.2	-3.1	2.4	6.4	11.0	16.6	22.4	21.1	15.6	8.2	2.3	-2.8		
	DBStd	10.47	6.26	6.90	6.43	5.33	5.02	4.49	3.50	3.87	5.52	6.02	6.84	6.79		
	HDD10.0	1928	378	366	243	131	49	4	0	0	15	104	241	397		
	HDD18.3	3998	636	599	494	358	232	84	8	19	113	317	480	656		
	CDD10.0	1282	0	1	7	22	80	202	383	345	184	48	10	1		
	CDD18.3	310	0	0	0	0	4	33	133	106	33	2	0	0		
	CDH23.3	3637	0	0	1	8	63	406	1526	1189	412	32	1	0		
	CDH26.7	1488	0	0	0	0	12	153	683	497	138	4	0	0		
(14) Wind	WSAvg	4.2	3.8	4.0	4.5	5.1	4.8	4.2	3.9	3.9	4.1	4.2	4.0	3.9		
(15) Precipitation	PrecAvg	382	12	12	23	43	67	60	40	19	29	35	16	10		
	PrecMax	486	30	30	46	101	163	131	116	48	60	100	44	24		
	PrecMin	223	1	1	4	16	4	15	6	0	3	6	4	2		
	PrecStd	68	8	6	11	22	43	31	26	13	15	24	9	6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.6	15.3	21.2	24.8	28.1	34.2	36.8	35.6	32.6	27.1	21.2	14.3		
		MCWB	4.8	6.0	8.3	10.9	13.8	16.0	16.4	15.7	14.6	11.4	8.4	4.9		
	2%	DB	10.3	11.5	17.4	21.5	24.9	30.8	34.2	33.0	30.2	23.2	17.5	11.1		
		MCWB	3.1	4.1	7.0	9.3	13.1	15.7	16.2	15.1	13.6	10.1	6.9	3.4		
	5%	DB	8.4	8.8	14.8	18.5	22.5	27.8	32.5	31.6	28.0	20.6	14.5	8.4		
		MCWB	2.2	2.6	5.7	8.3	12.4	15.4	16.0	15.1	12.9	9.4	5.5	2.0		
	10%	DB	6.4	6.3	12.0	15.9	20.0	25.2	30.8	29.5	25.7	17.5	12.0	6.2		
		MCWB	0.9	1.0	4.7	7.2	11.3	15.0	16.0	14.6	12.2	8.0	4.3	0.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.2	6.3	9.1	11.8	15.7	19.1	19.2	18.0	16.5	12.6	9.1	5.5		
		MCDB	12.5	14.2	19.6	23.1	23.9	27.0	29.4	28.5	27.1	22.8	19.9	13.6		
	2%	WB	3.6	4.4	7.4	9.9	14.1	17.6	18.3	17.2	14.9	11.1	7.3	3.7		
		MCDB	9.7	10.8	16.6	19.5	22.4	26.6	28.5	28.3	26.4	21.2	16.7	10.1		
	5%	WB	2.4	2.7	6.1	8.7	13.0	16.5	17.6	16.4	13.8	9.9	6.0	2.3		
		MCDB	7.9	8.4	13.9	17.3	21.2	25.9	28.1	27.3	25.4	19.1	14.1	8.0		
	10%	WB	1.1	1.3	4.8	7.5	11.9	15.3	16.8	15.6	12.8	8.6	4.6	1.0		
		MCDB	6.0	5.9	11.4	15.2	19.0	24.0	27.6	26.6	23.7	16.8	11.3	5.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.2	10.0	11.1	11.6	11.3	12.6	14.2	13.9	13.2	11.6	10.7	9.8		
		MCDBR	12.8	12.6	15.2	16.3	15.0	16.4	16.5	16.2	16.2	15.8	14.0	12.4		
	5% WB	MCWBR	7.9	7.5	7.8	7.9	7.3	7.1	5.9	5.9	6.5	7.4	7.5	7.5		
		MCWBR	12.4	11.9	14.2	15.0	13.7	14.7	14.5	14.3	14.5	14.4	13.3	11.8		
(40) Clear-Sky Solar Irradiance	taub		0.238	0.247	0.268	0.292	0.314	0.321	0.342	0.341	0.308	0.280	0.251	0.239		
	taud		2.559	2.550	2.542	2.487	2.489	2.494	2.465	2.445	2.533	2.578	2.570	2.551		
	Ebn at Noon		923	970	983	975	955	945	921	911	923	914	895	886		
	Edh at Noon		62	76	88	102	105	105	107	105	89	73	61	57		
(44) All-Sky Solar Radiation	RadAvg		1.86	2.79	4.02	5.03	5.92	7.06	7.18	6.15	4.82	3.13	2.05	1.54		
	RadStd		0.10	0.14	0.28	0.30	0.59	0.51	0.27	0.24	0.35	0.26	0.16	0.10		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page